

GHA as Nix CI

<https://github.com/stepbrobd>

Sponsorship disclosure



How does it work?

- Python wrapper
 - Reads config file and generate filter
 - Execute nix-eval-jobs
 - Some string mutation and print to stdout
- GHA matrix job dynamic dispatch
- Each job will be marked
 - Skipped?
 - Eval error?
 - Build with post-build-hook
 - Cache push finalize
 - Upload logs

How does it work?

```
$ nix run github:stepbrobd/atelier -- discover --workers 32 --systems x86_64-linux
```

```
::notice::Discovered 3 build cells in 1 chunk(s), 14 skipped
```

```
chunks=[{"name": "Build", "cells": "{\n  \"include\": [\n    {\n      \"system\": \"x86_64-linux\", \"runner\": \"ubuntu-latest\", \"label\": \"formatter.x86_64-linux\", \"installable\": \".#formatter.x86_64-linux\", \"error\": \"\", \"log\": \"log-formatter.x86_64-linux\", \"system\": \"x86_64-linux\", \"runner\": \"ubuntu-latest\", \"label\": \"legacyPackages.x86_64-linux.crates.deps\", \"installable\": \".#legacyPackages.x86_64-linux.crates.deps\", \"error\": \"\", \"log\": \"log-legacyPackages.x86_64-linux.crates.deps\", \"system\": \"x86_64-linux\", \"runner\": \"ubuntu-latest\", \"label\": \"legacyPackages.x86_64-linux.crates.howfastly-gen\", \"installable\": \".#legacyPackages.x86_64-linux.crates.howfastly-gen\", \"error\": \"\", \"log\": \"log-legacyPackages.x86_64-linux.crates.howfastly-gen\"}\n  ]\n}"]}]
```

```
skipped=[{"system": "x86_64-linux", "label": "checks.x86_64-linux.fmt", "reason": "Already in the binary cache"}, {"system": "x86_64-linux", "label": "checks.x86_64-linux.clippy", "reason": "Already in the binary cache"}, {"system": "x86_64-linux", "label": "checks.x86_64-linux.taplo", "reason": "Already in the binary cache"}, {"system": "x86_64-linux", "label": "checks.x86_64-linux.nixpkgs-fmt", "reason": "Already in the binary cache"}, {"system": "x86_64-linux", "label": "legacyPackages.x86_64-linux.crates.howfastly", "reason": "Already in the binary cache"}, {"system": "x86_64-linux", "label": "checks.x86_64-linux.test", "reason": "Already in the binary cache"}, {"system": "x86_64-linux", "label": "legacyPackages.x86_64-linux.howfastly", "reason": "Already in the binary cache"}, {"system": "x86_64-linux", "label": "checks.x86_64-linux.clippy-web", "reason": "Already in the binary cache"}, {"system": "x86_64-linux", "label": "devShells.x86_64-linux.default", "reason": "Already in the binary cache"}, {"system": "x86_64-linux", "label": "checks.x86_64-linux.e2e", "reason": "Already in the binary cache"}, {"system": "x86_64-linux", "label": "legacyPackages.x86_64-linux.crates.howfastly-compute", "reason": "Already in the binary cache"}, {"system": "x86_64-linux", "label": "checks.x86_64-linux.clippy-compute", "reason": "Already in the binary cache"}, {"system": "x86_64-linux", "label": "legacyPackages.x86_64-linux.serve", "reason": "Already in the binary cache"}, {"system": "", "label": "packages.*.*", "reason": "Excluded by rule"}]
```

What if you have a RLY BIG job

Matrix jobs can dispatch up to 200 variants

Chunking the matrix job ;)

<https://github.com/stepbrobd/rocquencourt/actions/runs/27853991727>

Cancel or timed out?

Dw we have a streaming cache push so
progresses are (partially) saved!

Configuration

```
systems = ["x86_64-linux"]
```

```
include = ["checks.*", "legacyPackages.**"]
```

```
exclude = ["packages.*"]
```

```
substitutes = ["ssh-ng://example.com?priority=10"]
```

Configuration

Native cache push

- Attic
- Cachix
- Nix3

Custom configuration

e.g. pipe operator

Or custom scripts for almost every step

- Pre install
- Post install

- Pre push
- Push command
- Post push

How to use this?

- \$1000/mo

or

- Provide compute for me

How to use this?

JK

How to use this?

- Public repo

And

jobs:

atelier:

uses: stepbrobd/atelier/.github/workflows/discover.yaml@master

Questions?

<https://github.com/stepbrobd/atelier>